Victor Chua

Singapore | +65 9190 5651 | chua1102@e.ntu.edu.sg | github.com/zeppyclown

Singaporean Citizen | linkedin.com/in/vicchua123 | Internship: May 11 – August 11 2027

AI Engineer | Production RAG Systems | Agentic AI & Multimodal Machine Learning

Summary

Computer Science undergraduate at Nanyang Technological University (Turing AI Scholars Programme) specializing in production RAG pipelines, agentic AI architecture, and multimodal machine learning. Demonstrated track record across 3 software and AI engineering internships, including architecting a production RAG system deployed to 24 mechanics and fine-tuning open-source vision models published on HuggingFace. Experienced in Python, Linux, QLoRA fine-tuning of open-weight models, and industrializing AI PoCs.

Technical Skills

AI & Agentic Systems
– **RAG**, **Agentic AI**, **LangGraph**, **ChromaDB**, Vector Databases, Prompt Engineering, LLM Inference
– Semantic Embeddings, Document Chunking, Structured Tool-Calling, Context-Aware Recommendations
– Open-Source AI Stacks, HuggingFace, **QLoRA / PEFT**, 4-bit Quantization, Unsloth, ONNX Export

Machine Learning & Multimodal AI
– **PyTorch**, **TwelveLabs**, MobileNetV3, Transfer Learning, Computer Vision, Optical Music Recognition
– TensorFlow, Scikit-learn, Feature Engineering, Failure Mode Analysis, Model Evaluation Metrics

Programming & Frameworks
– **Python**, **FastAPI**, JavaScript/TypeScript, Dart, Flutter, Node.js, SQL, REST API Design

Infrastructure & Cloud
– **Linux**, **Git**, AWS (S3, Cost Explorer), Firebase, Elasticsearch, SQLite, Colab GPU Training

Data Engineering & DevOps
– Grafana, Kafka, RabbitMQ, Amazon S3, Automated Monitoring, Distributed Pipelines, Data Reconciliation

Work & Research Experience

Production Automotive RAG System & Vector Search Pipeline — Dec 2025 – Jan 2026
Machine Learning Intern, Yap Motor Singapore

- Architected and deployed a production **RAG** system leveraging Google Gemini and **ChromaDB** vector embeddings to deliver context-aware automotive diagnostic recommendations to 24 mechanics daily.
- Optimized vector search and re-ranking algorithms with folder-based priority scoring, vehicle metadata extraction using regex NLP, and dynamic filter construction supporting multi-condition queries.
- Implemented intelligent document chunking and metadata sanitation pipelines with semantic vector embeddings, resolving ChromaDB schema validation and case-sensitive filtering edge cases.

On-Premises LLM Inference & Multimodal Document Automation — May 2026 – Aug 2026
AI & Software Engineer Intern, SUSS Digital Solutions & AI

- Built an on-premises **LLM** inference pipeline for privacy-sensitive university documents, enforcing strict data confidentiality while eliminating manual administrative processing delays.
- Automated document and video content categorization using multimodal analysis (**TwelveLabs**) and structured LLM prompts, accelerating internal review workflows across university departments.

Agentic AI Architecture & Published Multimodal OMR Model — Jun 2026 – Present
Creator & Developer, Drum Hub (AI Desktop Application)

- Designed a mixed-initiative **Agentic AI** architecture using **LangGraph** with explicit state machines, tool-calling schemas, and conversation memory to process complex inputs reliably.
- Trained a dual-head MobileNetV3 Optical Music Recognition model via two-phase transfer learning on a custom dataset; evaluated and published the trained model weights on **HuggingFace**.

Human-in-the-Loop Search-Based LLM Translation Prototype — May 2025 – Aug 2025
Research Assistant, NTU Turing AI Scholars Programme (TAISP)

- Engineered bidirectional generation logic for forward and backward LLM text suggestions, enabling real-time human-in-the-loop interactive translation workflows across complex datasets.

Enterprise Data Pipeline Monitoring & Bottleneck Optimization May 2023 – May 2024
Data Engineer Intern, DBS Bank Singapore

- Monitored large-scale data pipeline processing **3,000+** jobs daily; developed Grafana dashboards that identified bottlenecks, enabling engineering teams to achieve **20%** RAM reduction and **15%** faster execution.

Education

Nanyang Technological University (NTU) Aug 2024 – Dec 2028
Bachelor of Computing (Computer Science) with Turing AI Scholars Programme (TAISP)

- Advanced Coursework: AI Fundamentals, Calculus, Linear Algebra, Probability & Statistics

Leadership & Activities

TAISP x MLDA Workshop Lead, NTU School of Computing 2025
Led a Linear Regression and Neural Networks workshop for **207 participants**; designed end-to-end coding exercises covering practical ML pipelines.